

OBJECTIVE

ENSIMAG engineering student seeking a full-time permanent position starting in September 2027.

EDUCATION

- **ENSIMAG**
Engineering Degree in Computer Science
 - **Paris-Saclay University**
Bachelor's Degree in Computer Science
- Grenoble, France

Sep 2024 – Jan 2027

Paris, France

Sep 2022 – Jun 2024

EXPERIENCE

- **HCL Technologies**
Machine Learning Engineer Apprenticeship - HPE Telco Solutions acquisition
 - **Hewlett Packard Enterprise**
Machine Learning Engineer Apprenticeship
 - **Hewlett Packard Enterprise**
Research Engineer Apprenticeship - 6G-GOALS Project
- Grenoble, France

Aug 2026 – Present

Grenoble, France

Dec 2025 – Aug 2026

Grenoble, France

Sep 2024 – Dec 2025
- **SLM Fine-Tuning:** Built a domain-specific question-answering dataset from product documentation and fine-tuned SLMs.
 - **SLM Evaluation and Data Quality Assessment:** Evaluated SLMs outputs, dataset quality, and input formulation to assess response accuracy and improve models reliability.
 - **Technical & Business Presentations:** Presented technical findings, project progress, and business implications to management and engineering stakeholders.
 - **Graph based Root Cause Analysis:** Developed a graph-based root cause analysis approach to identify failure sources and trace dependencies across anomalous events using system logs.
 - **Model Deployment:** Deployed AI models in cloud native environments using Kubernetes and containerization.
 - **GPU Inference Optimization:** Profiled and optimized deep learning inference workloads on GPUs, reducing latency through model and runtime optimization techniques.
 - **Energy modeling:** Built datasets and trained machine learning models to predict inference energy consumption based on execution signatures.
 - **Reinforcement learning:** Designed a reinforcement learning agent to dynamically select GPU configurations and pruned model variants under resource constraints.

PUBLICATIONS

João H. I. de Souza, M. Merluzzi, **I. Djouab**, et al., “Semantic Communications with RIC-Centric AI-Driven Orchestration for 6G Networks,” *EuCNC & 6G Summit*, 2026.

PROJECTS

- **Kernel (ENSIMAG):** Design and incremental implementation of a minimal operating system kernel in C, running on an emulator.
- **Compiler (ENSIMAG):** Team-based development of a Java subset compiler (Deca), including a trigonometry extension.
- **CPU Design (ENSIMAG):** Designed a 16-bit CPU using VHDL.
- **Scientific Abstract Generator:** Implemented and trained a decoder only transformer language model from scratch to generate scientific abstracts.

SKILLS

- **Machine Learning:** Supervised, unsupervised, semi-supervised, reinforcement learning
- **Inference Optimization:** Quantization, pruning, sensitivity analysis, TensorRT, ONNX
- **Deep Learning:** Transformers, DNNs, CNNs, GNNs
- **LLM Systems:** RAG, agentic workflows, prompt engineering
- **Deployment & MLOps:** Kubernetes, Docker, MLflow, GitLab CI, Argo CD
- **Databases:** SQL, NoSQL, Vector databases, Data modeling
- **GPU Profiling:** NVML, CUDA
- **Programming:** C, Python/PyTorch, Java, x86-64 assembly, VHDL
- **Web:** HTML, CSS, JavaScript, React

ASSOCIATIVE ACTIVITIES

- **Asthmatic Children's Association:** Volunteer and assistant instructor supporting children with asthma through swimming activities.
- **Agoaré Paris-Saclay University:** Participated in student association activities and event organization.

LANGUAGES

- **Arabic:** Professional working proficiency
- **English:** Professional working proficiency
- **French:** Native
- **Italian:** Elementary proficiency